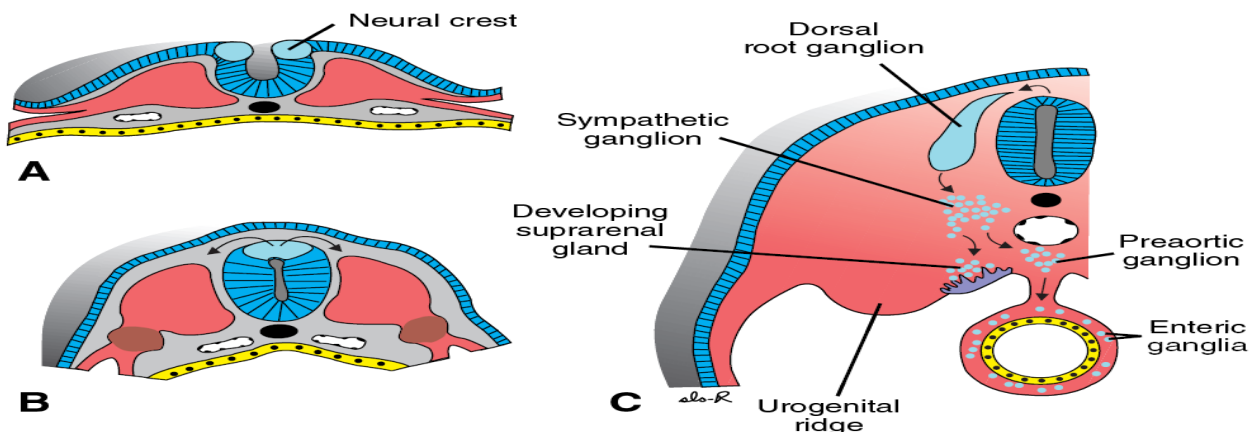


Development of the central nervous system

○ Formation of the neural tube (neurulation):

- Ectoderm thickens along dorsal midline to form the **neural plate**.
 - The neural plate invaginates, forming **a neural groove** flanked by **neural folds**.
 - Gradually, the neural folds approach each other in the midline, where they fuse.
 - Fusion begins in the cervical region and proceeds cranially and caudally.
 - As a result, the **neural tube** is formed. Neurulation is then complete, and the central nervous system is represented by a closed tubular structure with a narrow caudal portion, the **spinal cord**, and a much broader cephalic portion characterized by a few dilations, the **brain vesicles**.
- **Neural crest cells:** strip of ectodermal cells along the lateral edge of neural groove → fuse together → then separate as 2 longitudinal cords on each side of neural tube.
- neural crest derivatives
- Neurons of **dorsal root ganglia** of spinal nerves.
 - Neurons of **sensory ganglia** of cranial nerves **V, VII, VIII, IX, X**.
 - Neurons of **autonomic** (Sympathetic & Parasympathetic) ganglia.
 - **Schwann cells** of all peripheral nerves.
 - **Cells of adrenal medulla** (Chromaffin cells).
 - Melanoblasts of skin.
 - carotid body in the neck.
 - Arachnoid and pia mater

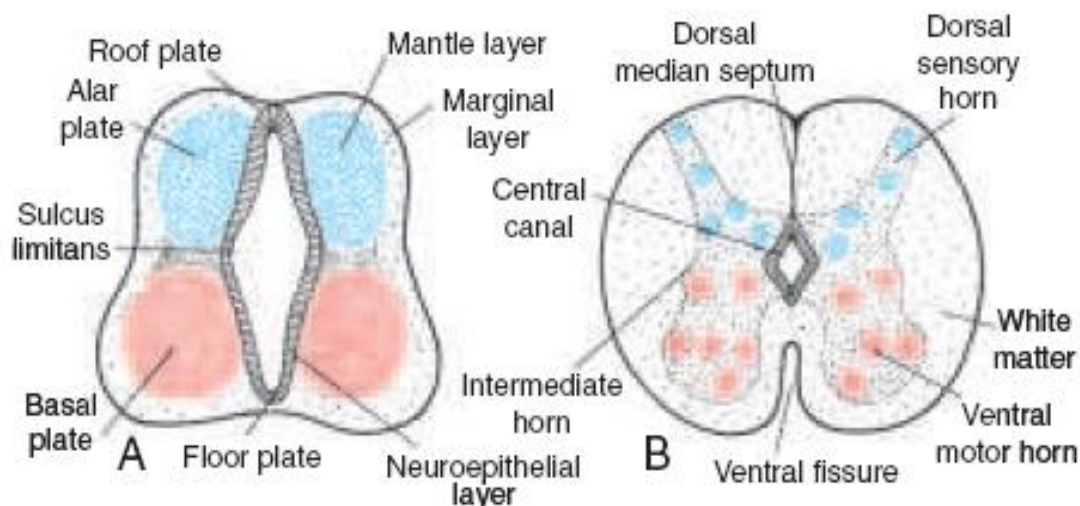


Formation and migration of neural crest cells in the spinal cord. **A, B.**

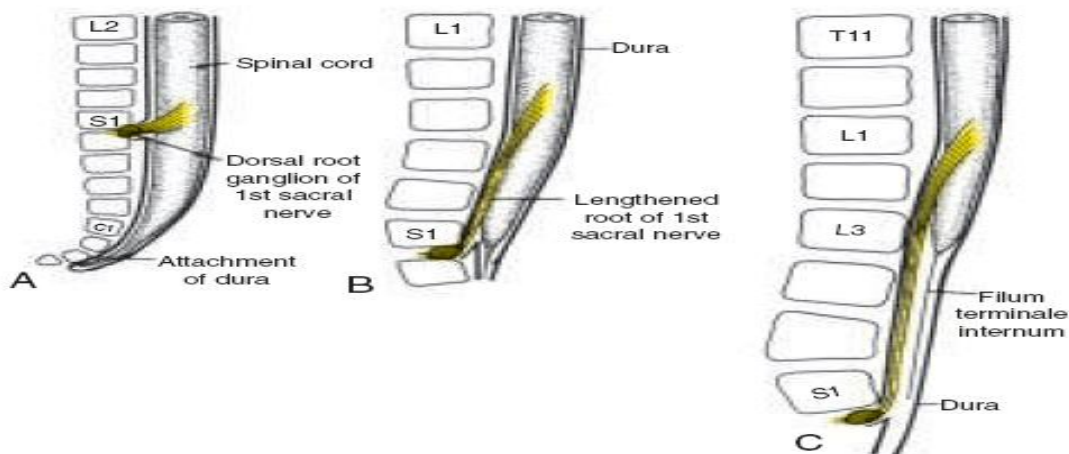
Crest cells form at the tips of neural folds and do not migrate away from this region until neural tube closure is complete.

Development of spinal cord:

- From the caudal part of neural tube:
- Differentiation of neuro-epithelium of neural tube:
 - **Ependymal layer:** lining the central canal.
 - Cells of the lateral wall divide rapidly producing large clusters of cells; **mantle layer** which will form the grey matter of spinal cord.
 - Differentiation of Mantle layer of neuro-epithelium into **Neuroblasts** and **Glioblasts** (Astroblast & oligodendroblast).
 - Nerve fibers of neuroblasts directed outwards; **marginal layer** which will form the white matter of spinal cord.
 - **Sulcus limitans** appear dividing **Proliferating mantle layer** of the lateral wall of the canal into:
 - Basal lamina (motor cells) → anterior horn.
 - **Proliferating mantle layer** will form a ventral thickening; **Basal lamina (AHCs,** motor cells) → anterior horn of spinal cord. Dorsal thickening: **Alar lamina (PHCs,** sensory cells) → posterior horn of spinal cord.
 - An **intermediate plate** is present to form **LHCs.**

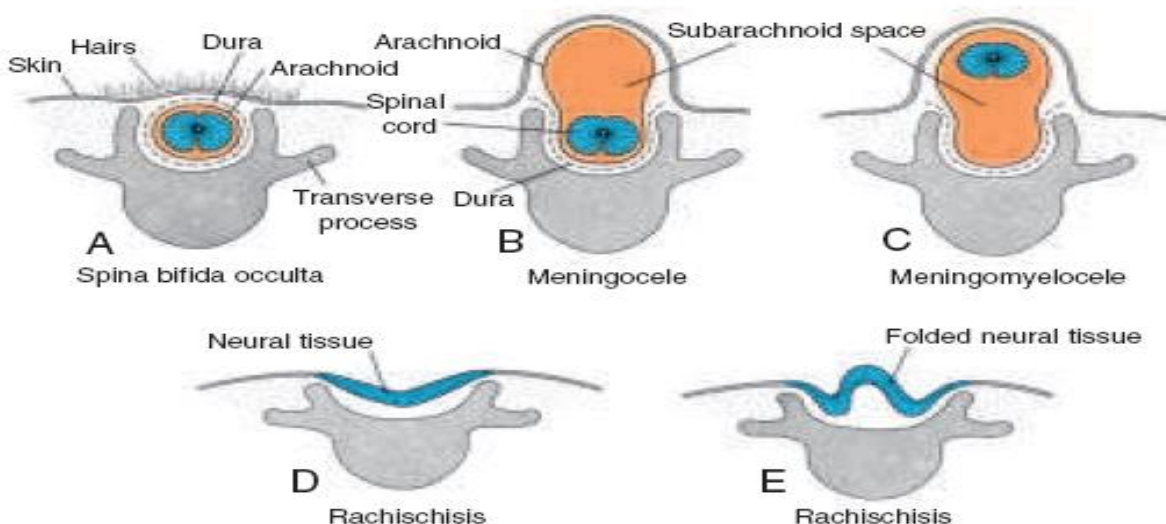


- **Positional changes of spinal cord:** due to faster growth of vertebral column than the neural tissue:
 - 3rd month fetus: spinal cord fills the whole length of vertebral canal.
 - 5th month fetus: lower end of spinal cord at the level of S1 vertebra.
 - Newborn: lower end of spinal cord at the level of L3 vertebra.
 - Adult: lower end of spinal cord at the level of disc between L1, 2 vertebrae.



- **Anomalies: (NTDs): Neural tube defects:**

- 1) **Spina bifida occulta:** failure of vertebral arch closure around spinal cord, common on lumbosacral region, covered by hairy skin.
- 2) **Spina bifida cystica:** associated with herniation of meninges (**Meningocele**), or with spinal nerves of cauda equina (**Meningo-myelocele**) or **Myelocele**: failure of closure of neural tube → spinal cord exposed.
- 3) **Rachischisis:** opened spinal cord which is continuous with skin surface.



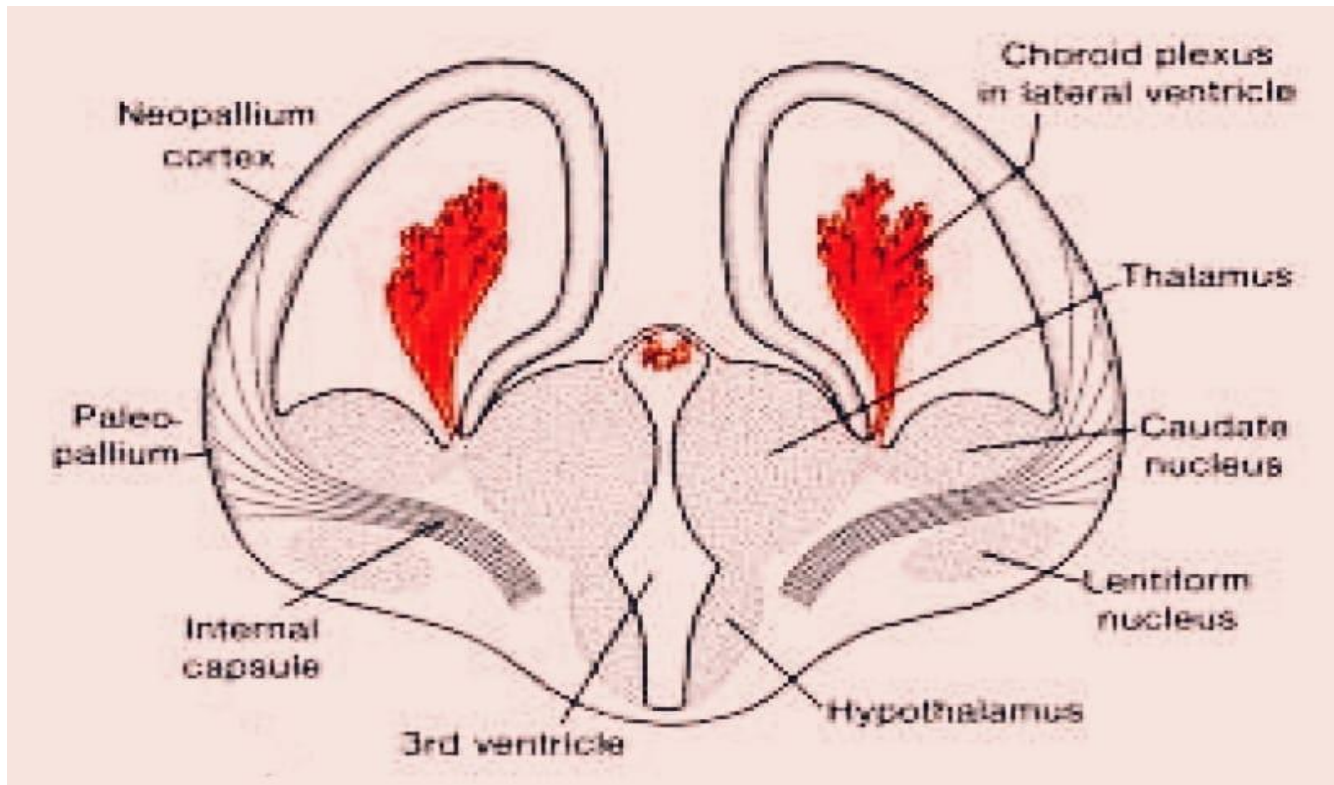
Development of THE BRAIN:

- The **cephalic end of the neural tube** shows **three dilations**, the primary brain vesicles: (a) the **prosencephalon**, or forebrain; (b) the **mesencephalon**, or midbrain; and (c) the **rhombencephalon**, or hindbrain.
- Simultaneously it forms three flexures: (a) the **cervical flexure** at the junction of the hindbrain and the spinal cord, (b) the **pontine flexure** divides the hindbrain and (c) the **cephalic flexure** in the midbrain region.

Derivatives of brain vesicles:

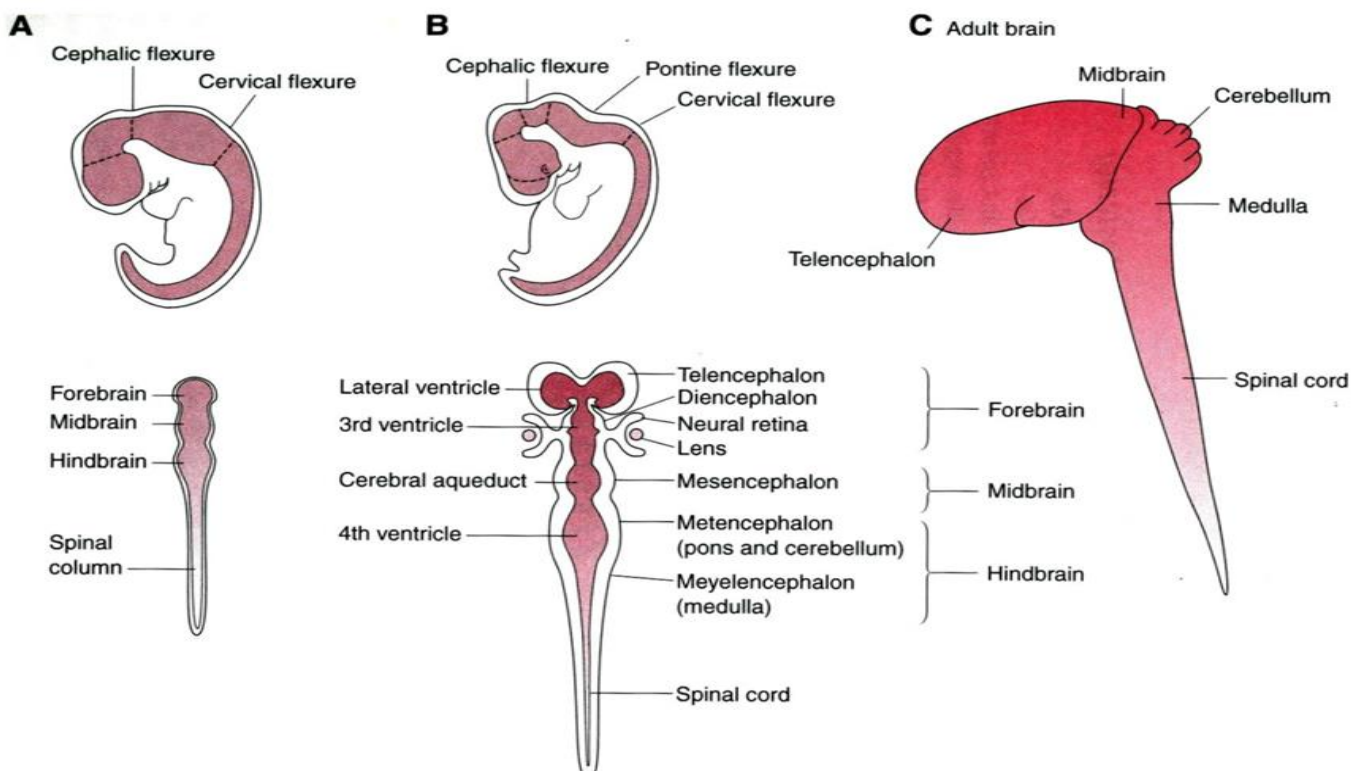
Vesicle	Derivatives		Cavity
Prosencephalon (forebrain)	Telencephalon	2 cerebral hemispheres	Lateral ventricle
	Diencephalon	optic cup and stalk pituitary, Thalamus, hypothalamus, epithalamus, metathalamic	3 rd ventricle
Mesencephalon (midbrain)	Midbrain		Cerebral aqueduct of Sylvius
Rhombencephalon (hindbrain)	Metencephalon	Pons – cerebellum	4 th ventricle
	Myelencephalon	Medulla oblongata	

- **Cerebral Hemispheres**: arise at the beginning of the fifth week of development as bilateral evaginations of the lateral wall of the prosencephalon.
- Continuous growth of the cerebral hemispheres in anterior, dorsal, and inferior directions results in the formation of frontal, temporal, and occipital lobes, respectively. As growth in the region overlying the corpus striatum slows, however, between the frontal and temporal lobes becomes depressed and is known as the **insula**.
- During the final part of fetal life, the surface of the cerebral hemisphere grows so rapidly that a great many convolutions (gyri) separated by fissures and sulci appear on its surface.
- **Cortex Development**: develops from the **pallium** which has two regions: (a) the **paleopallium**, immediately lateral to the corpus striatum and (b) the **neopallium**, between the hippocampus and the paleopallium.



Formation of cerebellum:

- From dorsolateral part of alar lamina of Metencephalon as **Rhombic lip**.
- **Alar lamina** of both sides bends medially then divide into medial and lateral bulges.
 - The 2 medial bulges → unite to form the **vermis**.
 - The 2 lateral bulges → unite to form **cerebellar Cortex**.



Congenital anomalies of the brain

- 1- **Hydrocephalus:** marked enlargement of skull due to accumulation of CSF in ventricular system due to blockage of CSF circulation.
- 2- **Anencephaly:** failure of brain development due to failure of closure of anterior neuropore. The cranial cavity is exposed with no skull cap.
- 3- **microcephaly:** small sized skull due to small sized brain.
- 4- **Meningocele: defect in the skull**→ meninges protrude.
- 5- **Meningo-encephalocele:** like previous but brain + meninges protrude.
- 6- **Cerebellar agenesis:** greater part of the cerebellum is absent.
- 7- **Faulty development of cerebral cortex:** may lead to congenital idiocy.
- 8- **Congenital absence of corpus callosum.**
- 9- **Holoprosencephaly:** Single telencephalic vesicle, no corpus callosum & fused eyes.